UNIVERSITY OF CAPE COAST

DEPARTMENT OF MATHEMATICS

TOPIC:

Understanding Invertibility Through Determinants

BY

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MAT 499: PROJECT WORK

A project report presented to the Department of Mathematics, Faculty of Science, University of Cape Coast in partial fulfillment of the requirements for the award of the Bachelor of Science Degree in Mathematics.

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Declaration

CANDIDATE'S DECLARATION

I, **Richmond Ntiako**, hereby declare that, except for references made to research works, this project is the result of my research and that no part of it has been presented for another degree in this university or elsewhere.

Richmond Ntiako	DATE

SUPERVISOR'S DECLARATION

I hereby declare that the preparation and presentation of the project work were supervised in accordance with the guidelines on supervision of project work laid down by the University of Cape Coast, Ghana.

Prof. Natalia Mensah	DATE

Abstract

This study investigates matrix invertibility with a focus on the determinant as the principal condition for determining whether a square matrix has an inverse. The research begins with fundamental concepts of matrices, operations, and transformations, then analyzes the determinant's mathematical properties and its link to rank, singularity, and inverse transformations. A descriptive methodology was employed, combining theoretical analysis, illustrative examples, and computational verification to demonstrate that a nonzero determinant is both a necessary and sufficient condition for invertibility. The findings emphasize that determinant-based evaluation is efficient for avoiding computational errors in solving systems of equations and has practical applications in cryptography, engineering, and numerical analysis. The study recommends integrating determinant concepts with real-world applications in mathematics education, encouraging the early use of computational tools, and expanding research into invertibility over more general algebraic structures.

Dedication

I dedicate this work to my beloved mother, **Grace Ntiako**, whose love, sacrifices, and unwavering support have sustained me. I also dedicate it to **my family** for their constant encouragement, to **my lecturers** whose teaching shaped my understanding of mathematics, and to my supervisor, **Prof. N. Mensah**, for her invaluable guidance and mentorship.

Acknowledgement

I am deeply thankful to the Almighty God for granting me the strength, wisdom, and determination necessary to bring this project to a successful conclusion. My sincere appreciation goes to my beloved mother, **Grace Ntiako**, whose unwavering love, sacrifices, and constant support have been a source of inspiration throughout my academic journey.
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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In linear algebra, the concept of matrix invertibility plays a crucial role in various mathematical and real-world applications. A square matrix is said to be invertible (non-singular) if there exists another matrix, called its inverse, such that their product results in the identity matrix. However, a key condition for a matrix to be invertible is that its determinant must be non-zero. The determinant serves as a mathematical tool that helps determine whether a matrix has an inverse, making it fundamental in areas like cryptography, engineering, computer science, and numerical analysis. Recent works reinforce the determinant's central role in invertibility. Pant et al. explored specific arithmetic patterns that influence determinant values, showing that structural regularities such as shared row divisibility can signal when a matrix will have a zero determinant. These observations strengthen the view that determinant behaviour is predictive of invertibility (Pant et al., 2024). Additionally, Danchev et al. (2020) provides broader mathematical context. While their studies examine matrix decompositions and arithmetic over rings respectively, their findings ultimately confirm that a non-zero determinant remains the definitive test for invertibility in standard linear algebra over fields. These works enrich the understanding of determinant mechanics, but do not contradict its primacy in determining whether a matrix is invertible. Therefore, this study focuses specifically on the determinant as the determining factor for matrix invertibility. It aims to clarify the theoretical underpinnings of this relationship and demonstrate how determinant-based criteria can be applied in practical computational settings, particularly in fields such as cryptography, where matrix invertibility is crucial for encoding and decoding operations. The determinant provides insight into many properties of a matrix, such as its rank, volume scaling factor, and singularity. If the determinant of a matrix is zero, the matrix is classified as singular and does not have an inverse, which can cause computational issues in solving systems of equations or performing transactions. By understanding the role of determinants in matrix invertibility, this research will provide theoretical insights and

practical computational approaches, demonstrating the importance of linear algebra in cryptography.

1.2 Problem Statement

In linear algebra, the concept of matrix invertibility is essential for solving systems of equations, modeling dynamic systems, and enabling computations in various applied fields such as cryptography, engineering, and economics. However, despite its importance, many students and practitioners lack a deep understanding of the precise mathematical condition that determines whether a matrix is invertible, namely, the role of the determinant.

A common challenge arises when working with square matrices: not all matrices have inverses, and failing to identify non-invertible (singular) matrices early can lead to computational errors, inefficiencies, and incorrect conclusions in both theoretical and practical applications. In many learning contexts, the determinant is treated as a mechanical procedure rather than a conceptual tool for understanding invertibility.

This project addresses the gap by thoroughly analyzing the determinant as the principal condition for matrix invertibility, emphasizing both its mathematical foundations and real-world significance. The study aims to clarify this relationship, explore its applications, and demonstrate how determinant-based techniques can be used to assess and interpret matrix invertibility effectively.

1.3 Objectives of the Study

The primary objective of this study is to analyze the relationship between determinants and matrix invertibility, emphasizing their significance in both theoretical mathematics and practical applications. To achieve this, the study will focus on the following specific objectives:

1. To examine the conditions for matrix invertibility by exploring the role of determinants in determining whether a matrix has an inverse.

2. To investigate the mathematical properties of determinants, including their impact on rank, volume scaling, and singularity.

3. To identify computational challenges associated with singular matrices, particularly in solving systems of equations and performing matrix transformations.

4. To explore real world applications of invertible matrices, with a focus on fields such as cryptography, engineering, and numerical analysis.

1.4 Significance of the Study

This study is significant in both theoretical and practical contexts. On a theoretical level, it deepens the understanding of matrix invertibility by highlighting the determinant as the principal criterion for determining whether a matrix is invertible. This insight is an essential concept in linear algebra that supports effective teaching and learning. Practically, the research is valuable to disciplines such as cryptography, computer science, and engineering, where matrix invertibility is crucial for tasks like encryption, modeling, and data processing. By focusing on determinant-based techniques, the study provides a reliable and efficient approach to identifying invertible matrices, thereby aiding students, educators, and professionals dealing with linear systems and computations.

1.5 Limitations

This study is limited in the following ways:

- It focuses solely on square matrices; non-square matrices are not considered.
- The determinant is examined as the primary and sufficient condition for invertibility; other advanced or abstract criteria (e.g., over non-commutative rings) are not deeply explored.
- The study assumes matrices over common fields such as $\mathbb{R}$ (real numbers) and $\mathbb{C}$ (complex numbers), limiting generalization to matrices over more abstract algebraic structures like rings or modules.
- Computational demonstrations are limited to basic examples; large-scale numerical simulations or matrix decompositions are beyond the scope.

1.6 Delimitations

This study is limited in the following ways:

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- Computational demonstrations are limited to basic examples; large-scale numerical simulations or matrix decompositions are beyond the scope.

1.7 Outline of the Study

This study is structured into five chapters, each focusing on a specific aspect of the investigation into the relationship between matrix invertibility and determinants.

Chapter One: Introduction

Background of the Study

This section introduces the concept of matrix invertibility, with particular emphasis on the role of the determinant in determining whether a matrix is invertible. It highlights the mathematical foundation of this relationship and its significance in solving systems of equations and addressing real-world problems in fields such as cryptography, engineering, and data science.

Objectives of the Study

The section outlines both the general and specific objectives of the study. It defines the primary aim of examining how determinants influence invertibility and articulates the specific research questions that guide the investigation.

Significance of the Study

Importance to theoretical mathematics, education, and applied fields like cryptography.

Scope and Limitations

Defines the focus on square matrices over fields, and any boundaries.

CHAPTER Two: Literature Review

Overview of Matrix Theory

This section presents a historical and conceptual background on matrix theory, including the development of matrices and determinants. It also introduces the formal definitions and foundational principles relevant to invertibility.

Review of Related Works

Here, existing scholarly literature is reviewed and critically analyzed to establish the context of the study. Contributions by prominent mathematicians are discussed to show how the current research builds on and fills gaps in previous work.

Chapter Three: Methodology

Research Design

The study employs a theoretical and mathematical research design, supplemented by computational demonstrations to validate concepts and findings.

Data Collection Techniques

Information and examples are gathered from established mathematical literature and through symbolic computation. The study is grounded in well-documented theoretical and empirical work.

Tools and Techniques

The study utilizes mathematical software tools and manual calculations to compute determinants and assess matrix invertibility.

Analytical Approach

A step-by-step analysis is conducted to evaluate determinant properties and their role in confirming or rejecting matrix invertibility.

Chapter Four: Analysis and Results

Case Studies of Matrices

This section presents selected examples of both invertible and non-invertible matrices, with detailed analysis based on determinant values.

Observations on Determinant Behaviour

Patterns in determinant behaviour are explored, including thresholds and exceptions that influence invertibility.

Discussion

The findings are interpreted considering the research objectives and compared with existing literature to draw logical conclusions.

Computational Demonstrations

Mathematical software is used to simulate and verify calculations, ensuring the accuracy and reliability of the results.

Chapter Five: Summary, Conclusion, and Recommendations

Summary of Findings

A concise synthesis of the major discoveries regarding the role of determinants in establishing matrix invertibility.

Conclusions

Final remarks are made regarding the implications of the findings and their contribution to mathematical theory and practice.

Recommendations

Practical suggestions are offered for educators, researchers, and professionals working in mathematics-related fields. Suggestions for Future Work Opportunities for further research are proposed, including the exploration of invertibility in more abstract algebraic structures such as rings and non-standard fields.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature on matrix theory, computational techniques, and their wide-ranging applications. It aims to trace the historical development of matrices and determinants, highlight significant theoretical contributions, and identify research gaps that justify the current study.

2.2 Historical Background

The evolution of matrix theory and determinant concepts spans centuries, with roots traced back to ancient civilizations. As early as 300 BC, the Babylonians studied linear problems involving simultaneous equations, evidenced by surviving clay tablets. The Chinese, between 200 BC and 100 BC, advanced closer to the matrix concept in the Nine Chapters on the Mathematical Art, solving problems now recognized as systems of linear equations using primitive matrix methods (O'Connor & Robertson, 1996). In the modern era, the term matrix was first coined by James Joseph Sylvester in 1850, who viewed it as an array from which determinants could be derived. Arthur Cayley, influenced by Sylvester, further formalized matrix theory. By 1853, Cayley had introduced the inverse of a matrix, and in 1858, he provided the first abstract definition of matrices, including operations such as addition, scalar multiplication, and inversion. Cayley also formulated what is now known as the Cayley-Hamilton theorem, suggesting that every square matrix satisfies its own characteristic equation, although he only formally verified it for smaller matrices (O'Connor & Robertson, 1996).

The foundation of determinant theory began with Gottfried Wilhelm Leibniz in the 17th century, who initially used it to solve linear systems. Cramer's Rule, developed in the 18th century, was a significant advancement, utilizing determinants explicitly to find unique solutions to systems of equations. Further refinements came from mathemati-

cians such as Bézout, Vandermonde, and Laplace, who introduced methods to calculate determinants and expanded their applications. Gauss contributed significantly to matrix and determinant theory through his work on quadratic forms and introduced ideas that evolved into Gaussian elimination (O'Connor & Robertson, 1996).
In the 19th century, Augustin-Louis Cauchy made landmark contributions by rigorously developing the theory of determinants, proving the multiplication theorem, and exploring minors, adjoints, and eigenvalues. He also laid the groundwork for matrix diagonalization and similarity. Mathematicians like Jacobi, Kronecker, and Weierstrass helped generalize determinant concepts through algorithmic and axiomatic approaches, pushing matrix theory closer to its modern form. By the 20th century, matrices and determinants had become central topics in linear algebra, thanks to works like Bôcher's Introduction to Higher Algebra and influential textbooks by Turnbull, Aitken, and Mirsky (O'Connor & Robertson, 1996).

2.3 Theoretical Background

Serre (2010) begins by defining a matrix as a rectangular array of elements arranged in rows and columns, emphasizing its role as a fundamental object in linear algebra. He discusses the notation and terminology associated with matrices, including the dimensions ($m \times n$) entries, and the various types of matrices such as square, diagonal, and identity matrices. The chapter delves into the operations that can be performed on matrices, such as addition, scalar multiplication, and matrix multiplication, highlighting their properties and the conditions under which these operations are defined. Serre also touches upon the concept of the transpose of a matrix and its significance. Furthermore, the chapter introduces the idea of matrices as representations of linear transformations, providing a bridge between abstract algebraic concepts and their concrete matrix counterparts. This perspective is crucial for understanding the applications of matrices in various areas of mathematics and science. Throughout the chapter, Serre emphasizes clarity and precision, aiming to build a solid foundation for readers to grasp more complex topics in subsequent chapters. His approach combines rigorous mathematical exposition with intuitive explanations, making the material accessible to readers with a basic understanding of linear algebra.
Shafarevich & Remizov on the other hand, provides a foundational exploration of matrices and determinants in the second chapter of Linear Algebra and Geometry. The chapter begins with the introduction of determinants for second and third order matrices, grounded in the context of solving linear algebraic systems. The authors then generalize this concept by defining determinants of arbitrary order through an inductive approach. Fundamental properties of determinants are examined, offering insights into their algebraic behaviour. A more abstract perspective follows, where determinants are rigorously

defined as antisymmetric multilinear functions of matrix rows. This conceptual shift is supported by elements of permutation theory, which aid in deriving explicit formulas for determinants. The chapter concludes with the definition of matrix rank and a detailed discussion of core matrix operations such as addition, multiplication, and inversion, along with their corresponding properties. This comprehensive treatment lays a strong theoretical foundation for further studies in linear algebra and its applications (Shafarevich & Remizov, 2012) . Silvester explores the properties of determinants in block matrices, particularly focusing on conditions under which the determinant of a block matrix can be expressed in terms of its sub-matrices. The study begins with an examination of 2×2 block matrices and extends to $m \times m$ block matrices, considering elements from a field or a commutative ring. A significant result presented is that if the blocks of matrix commute appropriately, the determinant of an entire matrix equals the determinant of a matrix formed by the determinants of its blocks. This finding has implications for simplifying determinant calculations in complex matrices and contribute to a deeper understanding of matrix theory (Silvester, 2000).

Chen and Hong (2020) delve into the divisibility properties of determinants of power matrices generated from integer-valued arithmetic functions, a topic situated at the intersection of number theory and matrix theory. Their work extends foundational contributions made in earlier studies which focused on the determinant properties of matrices derived from arithmetic functions like the Euler totient function, Möbius function, and divisor function.

Han & Li also investigates the conditions under which matrices over inclines are invertible and extends Cramer's Rule to this algebraic structure. Inclines are additively idempotent semi-rings where the product of two elements is less than or equal to either factor. This study generalizes existing results from Boolean and fuzzy matrices, providing a broader framework for understanding matrix invertibility in non-traditional algebraic systems (Han & Li, 2004). Allam explores the computational strategies for determining the determinant of square matrices of order n $\times$ n, a foundational operation in linear algebra. The study emphasizes both classical and computational approaches, aiming to improve efficiency for larger matrices where traditional cofactor expansion becomes computationally infeasible. Allam outlines the cofactor expansion method, suitable for small matrices, and then transitions to more scalable techniques like row reduction, LU decomposition, and recursive algorithms, which reduce computation time and complexity. The paper also discusses the importance of determinants in assessing matrix invertibility, solving systems of linear equations (via Cramer's Rule), and in applications such as eigenvalue problems and matrix theory. The work contributes to the field by offering insight into optimizing determinant calculations, particularly relevant in high-dimensional data analysis, engineering models, and numerical simulations.

2.4 Research Gap and Justification

While determinants are classically linked to invertibility, few studies deeply compare this with generalized inverses. Katz (2006) presents a significant contribution to the understanding of matrix invertibility by exploring the decomposition of infinite invertible matrices over division algebras, a context where traditional determinant-based methods may fail or become undefined. His work demonstrates that in infinite-dimensional and non-commutative settings, invertibility can be characterized through algebraic structures such as elementary, diagonal, and permutation-like matrices, rather than scalar-valued determinants. This decomposition approach broadens the conceptual framework of invertibility, emphasizing structural and group-theoretic properties over numerical criteria. For a project focused on "Understanding Invertibility Through Determinants", Katz's study provides an essential contrast, revealing the limitations of classical determinants and highlighting alternative methods for assessing invertibility in advanced algebraic systems. This underscores the need for a more generalized and robust understanding of invertibility that extends beyond conventional finite-dimensional linear algebra. Zhou et al. (2019) explore the core inverses of linear combinations of two core invertible matrices. Their study focuses on identifying the conditions under which the sum or difference of two core invertible matrices remains core invertible and provides explicit expressions for such inverses. This investigation is extended to dual core inverses, with further attention given to matrix structures and properties that support generalized invertibility. This study underscores the fundamental role of determinants in understanding matrix invertibility, particularly within the classical framework of square matrices. While modern research has introduced alternative approaches such as core and generalized inverses for handling singular or non-square matrices, this project specifically focuses on reaffirming and deepening our comprehension of invertibility through determinant-based methods. By examining the conditions under which a matrix is invertible solely through its determinant, the work reinforces the foundational connection between these two concepts and clarifies their applicability in both theoretical and practical contexts.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology adopted for the study "Understanding Invertibility Through Determinants." The methodology provides a systematic framework for analysing the role of determinants in establishing matrix invertibility. Given the mathematical and theoretical nature of the study, a purely analytical approach is adopted, complemented by computational techniques where appropriate. The chapter describes the research design, data sources, tools and techniques used for analysis, and the rationale for each methodological choice.

3.2 Research Design

The research is fundamentally theoretical and analytical, grounded in the principles of linear algebra. It employs a descriptive and explorative design, which allows for both the in-depth explanation of established concepts and the exploration of their boundaries and applications. The design is appropriate because the study aims to explain how and why determinants determine matrix invertibility, rather than to test hypotheses through empirical experimentation. The methodology also integrates elements of comparative analysis by examining the differences in determinant behaviour across various matrix types (e.g., square vs. singular, structured vs. random) and across different algebraic contexts (e.g., real numbers, inclines, and division algebras).

3.3 Data Collection Methods

Data for this research is secondary and conceptual, gathered from existing scholarly literature, textbooks, and journal articles on linear algebra, determinant theory, and matrix invertibility. Sources include classical references such as those by Horn, Serre, Shafarevich and Remizov, and Sylvester, and many more, as well as contemporary works on generalized inverses and computational methods. Additionally, worked examples and matrix case studies are used as data points. These examples are selected to illustrate key properties of determinants and to test their sufficiency in establishing matrix invertibility across various conditions.

3.4 Tools and Techniques

To ensure precision in the calculations and to handle large or complex matrices, mathematical software tools such as MATLAB, Mathematica, or Python (NumPy) are employed. These tools allow for the implementation of advanced techniques such as:

1. LU Decomposition
2. Row Reduction to Echelon Form
3. Cofactor Expansion
4. Symbolic Computation of Determinants
5. Automated Invertibility Testing
6. Manual calculations are also employed for simpler examples to illustrate fundamental principles and validate computational results.

3.5 Analytical Approach

The analytical process involves the step-by-step evaluation of the determinant of a matrix and its correlation with invertibility. The following procedure is adopted:

- Classification of Matrices: Square, singular, diagonal, block, and randomly generated matrices are categorized for comparative purposes.
- Determinant Evaluation: Determinants are computed manually and/or through software, and their values are analysed in relation to the matrix's rank and inverse.
- Invertibility Assessment: Based on determinant results, matrices are tested for invertibility. The relationship is evaluated using both classical (e.g., Cramer's Rule) and computational criteria.
- Pattern and Exception Analysis: The behaviour of determinants in edge cases (e.g., zero determinant, nearly singular matrices, or generalized inverse scenarios) is explored to highlight any limitations or exceptions in the classical rule.
- Comparative Discussion: Insights are drawn by comparing determinant-based invertibility with other techniques, such as the use of Moore–Penrose or core inverses, to demonstrate both the strengths and boundaries of determinant-based approaches.

3.6 Justification for the Methodology

The chosen methodology is justified by the theoretical focus of the research objective. Since the study deals with mathematical structures and properties, a qualitative or empirical design would not be suitable. The analytical framework provides the necessary rigour for exploring mathematical truths, while the inclusion of computational tools ensures relevance and accuracy in modern applications.
Furthermore, the methodology is aligned with the study's objective of reinforcing the foundational understanding of matrix invertibility through determinant analysis, offering both depth and clarity on the topic.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 MATRICES

4.1.1 Definition

A matrix is an m x n array of scalars from a field F. If $m = n$, the matrix is said to be ***square***.

The set of all m x n scalars over a field F is denoted as $M_{m,n}$ (F) and $M_{n,n}(F)$ respectively.

For example, if

$$A = \begin{bmatrix} 1 & 3 & \mathrm{i} \\ \pi & -2 & 0 \end{bmatrix} = a_{\mathrm{ij}} \qquad \ldots \qquad (4.1.1)$$

Then $A \in M_{2,3}(\mathbb{C})$ has entries $a_{11} = 1$, $a_{12} = 3$, $a_{13} = \mathrm{i}$, $a_{21} = \pi$, $a_{22} = -2$, and $a_{23} = -2$.

The above matrix A is considered a Complex matrix since the entries $a_{\mathrm{ij}} \in \mathbb{C}$, a Field of Complex.

(It is noticed that $\mathbb{R} \subset \mathbb{C}$, thus, the real numbers $\mathbb{R}$ is a subset of the complex numbers $\mathbb{C}$).

A ***submatrix*** of a given matrix is a rectangular array formed by selecting specific rows and columns from the original matrix.

Example:

$$\begin{bmatrix} \pi & -2 \end{bmatrix} \qquad \ldots \qquad (4.1.2)$$

The matrix (4.1.2) is a *submatrix* derived from row 2, column 1 and 2 of the matrix (4.1.1).

4.1.2 Order of matrices

The order of matrix can also be referred as the dimensions or size of the matrix.

If a matrix has:

- m rows and
- n columns,

Then the order is written as $m \times n$ which is read as "m by n". From example (4.1.1), the order of the matrix A is 2×3. If a square matrix has order 3, it implies the matrix has 3 rows and 3 columns represented as 3×3 respectively.

4.1.3 Types of Matrices

1. Row Matrix / Column Matrix

Definition: A **row matrix** is a type of matrix that has only one row and one or more columns. It is represented as a $1 \times n$ matrix, where n is the number of columns.
Example:

$$A = \begin{bmatrix} 1 & 3 & \pi \end{bmatrix}$$

This matrix has 1 row and 3 columns, and is represented as a 1×3 row matrix. In linear algebra, a row matrix is also called a *row vector*.
A **column matrix**, on the other hand, is a matrix with a single column and one or more rows. It is represented as an $n \times 1$ matrix, where n is the number of rows.
Example:

$$A = \begin{bmatrix} 1 \\ 3 \\ \mathrm{i} \end{bmatrix}$$

This matrix has 3 rows and 1 column, and is represented as a 3×1 column matrix. In linear algebra, it is called a *column vector*.

Properties:

- **Addition and Subtraction:** A row or column matrix can only be added to or subtracted from another matrix of the same size.

  Example:

$$\begin{bmatrix} 1 & 3 \end{bmatrix} + \begin{bmatrix} -2 & 0 \end{bmatrix} = \begin{bmatrix} 1+(-2) & 3+0 \end{bmatrix} = \begin{bmatrix} -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \end{bmatrix} - \begin{bmatrix} -2 & 0 \end{bmatrix} = \begin{bmatrix} 1-(-2) & 3-0 \end{bmatrix} = \begin{bmatrix} 3 & 3 \end{bmatrix}$$

- **Scalar Multiplication:** Each element of the matrix is multiplied by a scalar (a constant).

  Example:
  $$A = \begin{bmatrix} 1 & 3 & \mathrm{i} \end{bmatrix}, \quad 2A = 2 \cdot \begin{bmatrix} 1 & 3 & \mathrm{i} \end{bmatrix} = \begin{bmatrix} 2 & 6 & 2\mathrm{i} \end{bmatrix}$$

- **Matrix Multiplication:** A row matrix of size $1 \times n$ can be multiplied by a column matrix of size $n \times 1$, resulting in a scalar (a 1×1 matrix).

  Example:
  $$\begin{bmatrix} 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ \pi \end{bmatrix} = 1 \cdot 2 + 0 \cdot \pi = 2$$

2. Square Matrix

Definition:

A square matrix $n \,\mathrm{x}\, n$ is a matrix with the same number of rows and columns. It is an $n \,\mathrm{x}\, n$ matrix where n is a positive integer.
Example:
$$B = \begin{bmatrix} 1 & 3 \\ \pi & -2 \end{bmatrix}$$
This is a $2 \,\mathrm{x}\, 2$ square matrix because it has 2 rows and 2 columns.

Key Property:

Only square matrices can have;

- A determinant
- An inverse
- Eigenvalues and Eigenvectors

Types of Square Matrices

i. Diagonal Matrix: All off-diagonal elements are zero.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

ii. Identity Matrix: Diagonal elements are 1, and off-diagonal elements are 0.

$$I = \begin{bmatrix} 1 & 0 & ... & 0 \\ 0 & 1 & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

iii. Singular Matrix: Determinant $= 0$ (no inverse)

$$\begin{bmatrix} 4 & 6 \\ 2 & 3 \end{bmatrix}$$

iv. Nonsingular Matrix: Determinant $\neq 0$ (has an inverse).

$$\begin{bmatrix} 4 & 6 \\ 2 & -3 \end{bmatrix}$$

v. Symmetric Matrix: $A = A^T$ (transpose equals the original)

vi. Skew Matrix: $A = -A^T$

4.1.4 Matrix Operation

Definition:

A matrix operation is any mathematical procedure involving matrices. These concepts allow us to manipulate matrices to solve systems of equations, transform data, model real-world processes, and perform computations in various fields.

Matrix Addition

Matrix addition is the operation of adding two or more matrices of the same order by summing their corresponding entries. It is defined entrywise for arrays of the same dimensions and is denoted by $+$. Thus if $A = \left[a_{ij}\right]$ and $B = \left[b_{ij}\right]$ are both $m \times n$ matrices, then their sum $A + B = [a_{ij} + b_{ij}]$
is also an $m \times n$ matrix.
The ***zero matrix*** (all entries are zero) is the additive identity, and $M_{m,n}(F)$ is a vector space over F.
Example:

$$A = \begin{bmatrix} 9 & 4 & 10 \\ 8 & 5 & 9 \\ 7 & 6 & 7 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 4 & 8 \\ 5 & 9 & 6 \end{bmatrix}$$

Then $A + B$ implies

$$\begin{bmatrix} 9 & 4 & 10 \\ 8 & 5 & 9 \\ 7 & 6 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 7 & 4 & 8 \\ 5 & 9 & 6 \end{bmatrix} = \begin{bmatrix} 9+1 & 4+2 & 10+3 \\ 8+7 & 5+4 & 9+8 \\ 7+5 & 6+9 & 7+6 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 13 \\ 15 & 9 & 17 \\ 12 & 15 & 13 \end{bmatrix}$$

It is important to note that matrix addition is commutative. Thus $(A + B = B + A)$ respectively.

Theorem (4.1.4.1.1)

Let $P \subset M_{m,n}(F)$ and $Q \subset M_{m,n}(F)$ such that $P+Q = P$. Then we say Q is the additive identity given by:

$$Q = \begin{bmatrix} 0 & 0 & ... & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} = \mathbf{0}$$

Theorem (4.1.4.1.2)

Let $P \subset M_{m,n}(F)$ and $R \subset M_{m,n}(F)$ such that $P+R = Q$. Then we say R is the additive inverse of P given by: $R = -P$ where

$$Q = \begin{bmatrix} 0 & 0 & ... & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} = \mathbf{0}$$

Matrix Multiplication

Matrix multiplication is the process of multiplying two matrices to produce a new matrix. It is not done element-wise like the addition. Instead, the **dot product** of rows and columns is used. Matrix multiplication is denoted by juxtaposition ("AB") and corresponds to the composition of linear transformations. Therefore, it is defined only when $A \in M_{m,n}(\mathrm{F})$ and $B \in M_{n,q}(\mathrm{F})$. It is associative [thus $A(BC) = (AB)C$] but not always commutative.
For example,

$$\begin{bmatrix} 2 & 4 \\ 12 & 16 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ 6 & 16 \end{bmatrix}$$

The identity matrix

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \in M_n(\mathrm{F})$$

is the multiplication identity in $M_n(\mathrm{F})$; its main diagonals are 1, and all other entries are 0. The identity matrix and any scalar multiple of it (a *scalar matrix*) commute with every matrix in $M_n(\mathrm{F})$; they are the only matrices that do so. Matrix multiplication is distributive over matrix addition.

Thus, $A(B + C) = AB + AC$

Conditions for multiplication

You can multiply two matrices A and B only if:

- The number of columns in A = the number of rows in B.

  Thus, size A is $m \,\mathrm{x}\, n$ and size B is $n \,\mathrm{x}\, p$. Hence AB implies;

$$(m \,\mathrm{x}\, n\,).(n \,\mathrm{x}\, p) = m \,\mathrm{x}\, p$$

Example: Given

$$A = \begin{bmatrix} 1 & 3 & \mathrm{i} \\ \pi & -2 & 0 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 9 & 4 \\ 8 & 5 \\ 7 & 6 \end{bmatrix}$$

Then $(AB)_{ik} \Rightarrow \sum_{j=1}^{n} a_{\mathrm{i}\mathrm{j}} b_{k_{\mathrm{j}}}$

$$\begin{bmatrix} 1 & 3 & \mathrm{i} \\ \pi & -2 & 0 \end{bmatrix} \begin{bmatrix} 9 & 4 \\ 8 & 5 \\ 7 & 6 \end{bmatrix} = \begin{bmatrix} 9(1)+8(3)+7(\mathrm{i}) & 4(1)+5(3)+6(i) \\ 9(\pi)+8(-2)+7(0) & 4(\pi)+5(-2)+6(0) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 9+24+7\mathrm{i} & 4+15+6\mathrm{i} \\ 9\pi-16+0 & 4\pi-10+0 \end{bmatrix} = \begin{bmatrix} 33+7\mathrm{i} & 19+6\mathrm{i} \\ 9\pi-16 & 4\pi-10 \end{bmatrix}$$

4.1.5 Transpose of a Matrix

The transpose of a matrix is formed by flipping its rows into columns (or columns into rows).
If $A = [a_{ij}] \in M_{m,n}(\mathrm{F})$, the *transpose* of A, denoted by A^T, is the matrix in $M_{n,m}(\mathrm{F})$ whose i, j entry is $a_{\mathrm{j}\mathrm{i}}$; that is, rows are exchanged for columns and vice versa. For example,

$$A = \begin{bmatrix} 1 & 3 & \mathrm{i} \\ \pi & -2 & 0 \end{bmatrix}$$

then

$$A^T = \begin{bmatrix} 1 & 3 & \mathrm{i} \\ \pi & -2 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & \pi \\ 3 & -2 \\ \mathrm{i} & 0 \end{bmatrix}$$

Key Concepts

- $(A^T)^T = A$.

  Proof

  Let $A = [a_{ij}]$ be an $m \times n$ matrix. The transpose of A, denoted A^T, is defined as:

  $$(A^T)_{ij} = a_{ji}$$

  Now, take the transpose of A^T. That is,

  $$((A^T)^T)_{ij} = (A^T)_{ji}$$

  Substituting the definition of A^T, we get:

  $$((A^T)^T)_{ij} = a_{ij}$$

Since this holds for all i and j, it follows that:

$$(A^T)^T = A$$

- $(AB)^T = B^T A^T$

Proof

Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then, the product AB is an $m \times p$ matrix.

The (i, j)-th entry of AB is given by:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Now consider the transpose of AB, denoted $(AB)^T$. The (i, j)-th entry of $(AB)^T$ is the (j, i)-th entry of AB, so:

$$((AB)^T)_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

Now consider B^T and A^T.

B^T is a $p \times n$ matrix with entries $(B^T)_{ik} = b_{ki}$
A^T is a $n \times m$ matrix with entries $(A^T)_{kj} = a_{jk}$

The $(1, j)^{t^h}$ entry of the product $B^T A^T$ is:

$$(B^T A^T)_{ij} = \sum_{k=1}^{n} (B^T)_{ik} (A^T)_{kj} = \sum_{k=1}^{n} b_{ki} a_{jk}$$

This matches the expression for $((AB)^T)_{ij}$, so we conclude:

$$(AB)^T = B^T A^T$$

4.1.6 Conjugate Transpose (Hermitian Transpose)

The conjugate transpose of a matrix A, denoted by A^* is the matrix obtained by transposing A, and taking the complex conjugate of each entry. Notation: $A^* = \overline{A}^T$. (taking the complex conjuage, denoted by $\overline{A}$ then taking the transpose). Example: Let

$$A = \begin{bmatrix} 2 & i \\ 2-i & 6 \end{bmatrix}$$

The conjugate transpose is given by:

$$A^* = \begin{bmatrix} 2 & 2+i \\ -i & 6 \end{bmatrix}$$

Key concept

The conjugate transpose also obeys the *reverse order law*: $(AB)^* = B^*A^*$

Trace of a matrix

The trace of a square matrix A, denoted by $tr(A)$, is the sum of the entries on its main diagonal.
Given $A = [a_{ij}] \in M_{m,n}(\mathrm{F})$,

$$trA = \sum_{i=1}^{n} a_{ii} = a_{11} + ... + a_{qq}$$

in which $q = \min(m, n)$.

Key concept

- For any $A = [a_{ij}] \in M_{m,n}(\mathrm{C})$, $\mathrm{tr}AA^* = trA^*A = \sum_{i,j=1} |a_{ij}|^2$, so $trAA^* = 0$ if and only if $A = 0$

Proof

Let $A = [a_{ij}] \in M_{m,n}(\mathbb{C})$. Then,

$$\mathrm{tr}(AA^*) = \mathrm{tr}(A^*A) = \sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2$$

and therefore,

$$\mathrm{tr}(AA^*) = 0 \iff A = 0$$

Recall that for a matrix $A = [a_{ij}]$, its conjugate transpose is $A^* = [\overline{a_{ji}}]$. The $(i,i)^{th}$ entry of (AA^*) is given by:

$$(AA^*)_{ii} = \sum_{k=1}^{n} a_{ik}\overline{a_{ik}} = \sum_{k=1}^{n} |a_{ik}|^2$$

Therefore, the trace is:

$$\text{tr}(AA^*) = \sum_{i=1}^{m}(AA^*)_{ii} = \sum_{i=1}^{m}\sum_{k=1}^{n} |a_{ik}|^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2$$

Since $|a_{ij}|^2 \geq 0$ for all i, j, we have:

$$\text{tr}(AA^*) = 0 \iff |a_{ij}|^2 = 0 \text{ for all } i,j \iff a_{ij} = 0 \text{ for all } i,j \iff A = 0$$

4.1.7 Linear Transformation

Matrix can also be defined as a representation of a linear transformation of vectors.
Any matrix $A \in R^{m \times n}$ defines a linear transformation:

$$T : R^n \mapsto R^m$$

where $T(x) = A(x)$
Let U be an $n - dimensional$ vector space and let V be $m - dimensional$ vector space, both over the same field F; Let e be a basis of U and f be a basis of V. The isomorphisms $x \to x_e$ and $y \to y_f$ to represent vectors in U and V as $n - vectors$ and $m - vectors$ over F, respectively.
A linear transformation is a function $T : U \mapsto V$ such that

$$T(a_1x_1 + a_2x_2) = a_1T(x_1) + a_2T(x_2)$$

for any scalars a_1, a_2 and vectors x_1, x_2.
A matrix $A \in M_{m,n}(F)$ corresponds to a linear transformation $T : U \mapsto V$ in the following way: $y = T(x)$ if and only if $y_f = Ax_e$. The matrix A is said to represent the linear transformation T(relative to the base e and f); the representing matrix A depends on the bases chosen. When we study a matrix A, we realize that we are studying a linear transformation relative to a particular choice of bases, but explicitly appeal to the bases is usually not necessary.

4.1.8 Column and Row Space

The **column space**(range) of an $m \times n$ matrix A is the subspace of R^m spanned by the columns of A.

$$Col(A) = span\{x_1, x_2, ..., x_n\}$$

The dimension of the column space is the **rank** of the matrix. The column space tells us the set of all possible outputs (range) for the matrix transformation Ax.
The **row space** of an $m \times n$ matrix A is the subspace of R^n spanned by the rows of A.

$$Row(A) = span\{rowsofA\}$$

The dimension of the row space is also the **rank** of the matrix. The row space represents all possible linear constraints imposed on solutions to $Ax = b$.

Computations

1. Consider the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 1: Eliminate the entry below the leading 1 in the first column.

$$R_2 \to R_2 - 2R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now the matrix is in row echelon form because:

- Each leading entry is to the right of the one in the row above,
- All nonzero rows are above rows of zeros,
- Entries below each pivot are zero.

So, the row echelon form of A is:

$$\text{REF}(A) = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Row Space: The row space is the subspace of $\mathbb{R}^3$ spanned by the nonzero rows of the row echelon form of A:

$$\text{Row Space}(A) = \text{span}\left\{\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\right\}$$

Column Space: To find the column space, we identify the pivot columns from the row echelon form. These correspond to columns 1 and 3 in the original matrix. So the column space is:

$$\text{Col Space}(A) = \text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 1 \end{bmatrix}\right\}$$

2. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 1 & 2 & 1 \\ 1 & 2 & 4 & 6 & 6 \\ 0 & 0 & 0 & 1 & 1 \\ 2 & 4 & 6 & 8 & 10 \end{bmatrix}$$

We now row-reduce the matrix to row echelon form.

Step 1: Eliminate below Row 1

$$R_3 \to R_3 - R_1, \quad R_5 \to R_5 - 2R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This is the row echelon form of A.

Row Space: The row space is spanned by the non-zero rows in REF:

$$\text{Row Space}(A) = \text{span}\left\{ \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \end{bmatrix} \right\}$$

Column Space: From the REF, pivots appear in columns 1, 2, 3, and 4.

We extract the corresponding columns from the original matrix:

$$\text{Col Space}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 4 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 6 \\ 1 \\ 8 \end{bmatrix} \right\}$$

4.2 DETERMINANTS

4.2.1 Definition:

Determinants are scalar values associated with square matrices. Its domain is $M_n(F)$, and it may be presented in several different ways. We denote the determinant of $A \in M_n(F)$ by $det\, A$.

4.2.2 General Formula (Leibniz Formula)

$$detA = \sum_{\sigma \in S_n} \varepsilon(\sigma) \cdot \pi_\sigma(A)$$

where

$$\pi_\sigma(A) = a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

- Let $A = \left[a_{ij}\right]$
- $\sigma \in S_n$ is a permutation of the numbers $1, 2, \cdots, n$.
- $\pi_\sigma(A)$ means:

  Take one element from each row of A, and from column $\sigma(i)$ in row i, then multiply them all together.

- $\varepsilon(\sigma) \in \{+1, -1\}$ is the sign of permutation σ; $+1$ for even permutations (even number of swaps), and -1 for odd permutations (odd number of swaps).
  Thus
  $$\varepsilon(\sigma) = \prod_{i=1}^{n} \frac{\sigma(j) - \sigma(i)}{j - i}$$

Using the General Formula to prove matrices of order 2 and 3

I. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $n = 2$ There are 2 permutations in S_2: (σ_1, σ_2)
We are working with the set {1,2}, so all permutation σ of size 2 are;

- $\sigma_1 = (1, 2)$, $\sigma(1) = 1$ and $\sigma(2) = 2$

$$\varepsilon(\sigma) = \prod_{i=1}^{2} \frac{\sigma(j) - \sigma(i)}{j - i}$$

$$\Rightarrow \varepsilon(\sigma_1) = \prod_{i=1}^{2} \frac{\sigma(1) - \sigma(2)}{1 - 2} = \frac{1 - 2}{1 - 2} = \frac{-1}{-1} = +1$$

Hence $\varepsilon(\sigma_1) = +1$

- $\sigma_2 = (2,1), \qquad \sigma(1) = 2 \text{ and } \sigma(2) = 1$

$$\Rightarrow \varepsilon(\sigma_2) = \prod_{i=1}^{2} \frac{\sigma(1) - \sigma(2)}{1-2} = \frac{2-1}{1-2} = \frac{1}{-1} = -1$$

Hence $\varepsilon(\sigma_2) = -1$

Now

$$detA = \sum_{\sigma \in S_2} \varepsilon(\sigma) \cdot \pi_\sigma(A)$$

where

$$\pi_\sigma(A) = a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

$$det(A) = \sum_{\sigma \in S_2} \varepsilon(\sigma_1) \cdot a_{1\sigma_1(1)} a_{2\sigma_1(2)} + \varepsilon(\sigma_2) \cdot a_{1\sigma_2(1)} a_{2\sigma_2(2)}$$

$$= (+1) \cdot a_{11}a_{22} + (-1) \cdot a_{12}a_{21}$$

$$= a_{11}a_{22} - a_{12}a_{21}$$

$$= ad - bc$$

II. Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ $\qquad n = 3 \qquad S_3 = 3! = 6$ permutations: $(\sigma_1, \sigma_2, \cdots, \sigma_6)$

Using the formula

$$\varepsilon(\sigma) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i}$$

to find the sign $\varepsilon(\sigma)$ of permutation in S_3.

Using the set {1,2,3}, so all permutation σ of size 3 are;

- $\sigma_1 = (1,2,3)$, the identity permutation.

$$\sigma_1(1) = 1, \qquad \sigma_1(2) = 2, \qquad \sigma_1(3) = 3$$

$\varepsilon(\sigma_1) = +1$

- $\sigma_2 = (1,2) = (2,1,3)$

$$\sigma_2(1) = 2, \qquad \sigma_2(2) = 1, \qquad \sigma_2(3) = 3$$

$\varepsilon(\sigma_2) = -1$

- $\sigma_3 = (1,3) = (3,2,1)$

$$\sigma_3(1) = 3, \qquad \sigma_3(2) = 2, \qquad \sigma_3(3) = 1$$

  $\varepsilon(\sigma_3) = -1$
- $\sigma_4 = (2,3) = (1,3,2)$

$$\sigma_4(1) = 1, \qquad \sigma_4(2) = 3, \qquad \sigma_4(3) = 2$$

  $\varepsilon(\sigma_4) = -1$
- $\sigma_5 = (2,3,1)$
  $\varepsilon(\sigma_5) = +1$
- $\sigma_6 = (3,1,2)$
  $\varepsilon(\sigma_6) = +1$

To generate the terms,

- $\sigma_1 = (1,2,3) = a_{11}a_{22}a_{33}$
- $\sigma_2 = (1,2) = (2,1,3) = a_{12}a_{21}a_{33}$
- $\sigma_3 = (1,3) = (3,2,1) = a_{13}a_{22}a_{31}$
- $\sigma_4 = (2,3) = (1,3,2) = a_{11}a_{23}a_{32}$
- $\sigma_5 = (2,3,1) = a_{12}a_{23}a_{31}$
- $\sigma_6 = (3,1,2) = a_{13}a_{21}a_{32}$

Now

$$det(A) = \sum_{\sigma \in S_3} \varepsilon(\sigma_1) \cdot a_{1\sigma_1(1)}a_{2\sigma_1(2)}a_{3\sigma_1(3)} + \varepsilon(\sigma_2) \cdot a_{1\sigma_2(1)}a_{2\sigma_2(2)}a_{3\sigma_1(3)} + \cdots$$

$$+\varepsilon(\sigma_6) \cdot a_{1\sigma_6(1)}a_{2\sigma_6(2)}a_{3\sigma_6(3)}$$

$$= (+1) \cdot a_{11}a_{22}a_{33} + (-1) \cdot a_{12}a_{21}a_{33} + (-1) \cdot a_{13}a_{22}a_{31} + (-1) \cdot a_{11}a_{23}a_{32} +$$

$$(+1) \cdot a_{12}a_{23}a_{31} + (+1) \cdot a_{13}a_{21}a_{32}$$

$$= a_{11}a_{22}a_{33} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$

$$= aei - bdi - ceg - afh + bfg + cdh$$

$$= aei + bfg + cdh - afh - bdi - ceg$$

4.2.3 Properties of Determinants

1. Multilinearity property in rows
2. Alternating property
3. Normalization

Multilinearity

A function is multilinear if it's linear in each row separately while keeping the other rows fixed. So for the determinant function det: $\mathbb{R}^{nxn} \to \mathbb{R}$, the following are true:

- Linearity in each row
  Let A be a matrix with rows $r_1, \cdots, r_i, \cdots, r_n$. If row i is a linear combination of two vectors:
  $$r_i = a \cdot r_i' + b \cdot r_i''$$
  Then:
  $$\det(r_1, \ldots, r_i, \ldots, r_n) = a \cdot \det(r_1, \ldots, r_i', \ldots, r_n) + b \cdot \det(r_1, \ldots, r_i'', \ldots, r_n)$$

- Scalability in a row
  If you multiply any one row of a matrix by a scalar λ:
  $$\det(\cdots, \lambda r_i, \cdots) = \lambda \cdot \det(\cdots, r_i, \cdots)$$

- Additivity in a row
  If one row is the sum of two vectors, the determinant is the sum of the determinants:
  $\det(\cdots, r_{i'} + r_{i'}'', \cdots) = \det(\cdots, r_{i'}, \cdots) + \det(\cdots, r_{i'}'', \cdots)$

Proof

Let row i of matrix A be $r_i = a \cdot r_i' + b \cdot r_i''$.

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \left(\prod_{\substack{j=1 \\ j \neq i}}^{n} a_{j\sigma(j)} \right) (a \cdot a'_{i\sigma(i)} + b \cdot a''_{i\sigma(i)})$$

$$= a \cdot \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \left(\prod_{\substack{j=1 \\ j \neq i}}^{n} a_{j\sigma(j)} \cdot a'_{i\sigma(i)} \right) + b \cdot \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \left(\prod_{\substack{j=1 \\ j \neq i}}^{n} a_{j\sigma(j)} \cdot a''_{i\sigma(i)} \right)$$

$$= a \cdot \det(A') + b \cdot \det(A'')$$

Example:

Let's define matrix A as:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

We'll now express row 2 as a linear combination:

$$r_2 = 2 \cdot r_2' + 3 \cdot r_2''$$

where:

- $r_2' = \begin{bmatrix} 1 & 0 \end{bmatrix}$
- $r_2'' = \begin{bmatrix} 0 & 1 \end{bmatrix}$

So:

$$r_2 = 2 \cdot \begin{bmatrix} 1 & 0 \end{bmatrix} + 3 \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix}$$

Now define two new matrices:

$$A' = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \quad A'' = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Then: $A = 2A' + 3A''$

$$\text{Since } r_2 = 2 \cdot r_2' + 3 \cdot r_2'', \text{ then:}$$

$$\det(A) = 2 \cdot \det(A') + 3 \cdot \det(A'') = 2(-2) + 3(1) = -1$$

$$\text{Direct computation: } \det\left(\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}\right) = (1)(3) - (2)(2) = -1$$

Alternating property

Let $A = [a_{ij}]$ be an $n \times n$ matrix. The determinant of A is defined by the general (Leibniz) formula as:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot a_{1\sigma(1)} \cdot a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

where:

- S_n is the set of all permutations of $\{1, 2, \ldots, n\}$,
- $\operatorname{sgn}(\sigma)$ denotes the sign of the permutation σ.

Suppose two rows of A are equal, say row i and row j with $i \neq j$. This implies:

$$a_{ik} = a_{jk} \quad \text{for all } k = 1, 2, \ldots, n.$$

Now, for any permutation $\sigma \in S_n$, consider the term:

$$\text{sgn}(\sigma) \cdot a_{1\sigma(1)} \cdots a_{i\sigma(i)} \cdots a_{j\sigma(j)} \cdots a_{n\sigma(n)}.$$

Let τ be the transposition that swaps i and j, and consider the permutation $\tau \circ \sigma$. Then:

$$\text{sgn}(\tau \circ \sigma) = -\text{sgn}(\sigma),$$

and since rows i and j are equal, the products:

$$a_{i\sigma(i)} = a_{j\sigma(i)} \quad \text{and} \quad a_{j\sigma(j)} = a_{i\sigma(j)}.$$

Therefore, the terms corresponding to σ and $\tau \circ \sigma$ in the determinant expansion are equal in magnitude but opposite in sign:

$$\text{sgn}(\sigma) \cdot P + \text{sgn}(\tau \circ \sigma) \cdot P = P - P = 0,$$

where P represents the product of the matrix entries in that term. Since each term in the summation can be paired in this way and cancelled, the entire sum evaluates to zero:

$$\Rightarrow \det(A) = 0.$$

Conclusion: If a matrix has two equal rows, its determinant is zero. This verifies the alternating property.

Example:

Consider the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 4 & 5 & 6 \end{bmatrix}$$

Note that rows 2 and 3 are equal. We compute the determinant using the general formula:

$$\det(A) = \sum_{\sigma \in S_3} \text{sgn}(\sigma) \cdot a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)}$$

The six permutations in S_3 and their signs are:

$$\begin{aligned}
\sigma_1 &= (1,2,3), \quad \text{sgn} = +1 \\
\sigma_2 &= (1,3,2), \quad \text{sgn} = -1 \\
\sigma_3 &= (2,1,3), \quad \text{sgn} = -1 \\
\sigma_4 &= (2,3,1), \quad \text{sgn} = +1 \\
\sigma_5 &= (3,1,2), \quad \text{sgn} = +1 \\
\sigma_6 &= (3,2,1), \quad \text{sgn} = -1
\end{aligned}$$

Now we compute each term:

$$\begin{aligned}
\det(A) &= (+1)(1)(5)(6) + (-1)(1)(6)(5) + (-1)(2)(4)(6) + \\
&\quad (+1)(2)(6)(4) + (+1)(3)(4)(5) + (-1)(3)(5)(4) \\
&= 30 - 30 - 48 + 48 + 60 - 60 \\
&= 0
\end{aligned}$$

Normalization

The determinant of the identity matrix is equal to 1:

$$\det\left(I_n\right) = 1$$

for any identity matrix $\left(I_n\right) \in \mathbb{R}^{nxn}$.

Proof

Let I_n be the $n \times n$ identity matrix. Its entries are defined as:

$$a_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Using the general determinant formula:

$$\det(I_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

In the identity matrix, the only non-zero term in the sum corresponds to the identity permutation (id) where $\sigma(i) = i$ for all i:

$$\det(I_n) = \operatorname{sgn}(\mathrm{id}) \cdot a_{11} a_{22} \cdots a_{nn}$$

But

- $\operatorname{sgn}(\mathrm{id}) = +1$
- $a_{ii} = 1$ for all i

So:

$$\det(I_n) = (+1) \cdot 1 \cdot 1 \cdot 1 \cdots 1 = 1$$

All other terms in the sum involve at least one off-diagonal entry $a_{ij} = 0$, and hence contribute zero. Therefore:

$$\boxed{\det(I_n) = 1}$$

Example:

Let us consider the 3×3 identity matrix:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We apply the general determinant formula:

$$\det(I_3) = \sum_{\sigma \in S_3} \operatorname{sgn}(\sigma) \cdot a_{1\sigma(1)} \cdot a_{2\sigma(2)} \cdot a_{3\sigma(3)}$$

There are $3! = 6$ permutations in S_3:

- $\sigma_1 = (1, 2, 3), \quad \operatorname{sgn} = +1$
- $\sigma_2 = (1, 3, 2), \quad \operatorname{sgn} = -1$
- $\sigma_3 = (2, 1, 3), \quad \operatorname{sgn} = -1$
- $\sigma_4 = (2, 3, 1), \quad \operatorname{sgn} = +1$
- $\sigma_5 = (3, 1, 2), \quad \operatorname{sgn} = +1$
- $\sigma_6 = (3, 2, 1), \quad \operatorname{sgn} = -1$

We now evaluate each term:

$$\begin{aligned} \det(I_3) &= (+1)(a_{1,1})(a_{2,2})(a_{3,3}) + (-1)(a_{1,1})(a_{2,3})(a_{3,2}) + \\ &\quad (-1)(a_{1,2})(a_{2,1})(a_{3,3}) + (+1)(a_{1,2})(a_{2,3})(a_{3,1}) + \\ &\quad (+1)(a_{1,3})(a_{2,1})(a_{3,2}) + (-1)(a_{1,3})(a_{2,2})(a_{3,1}) \\ &= (+1)(1)(1)(1) + (-1)(1)(0)(0) + (-1)(0)(0)(1) \\ &\quad + (+1)(0)(0)(0) + (+1)(0)(0)(0) + (-1)(0)(1)(0) \\ &= 1 + 0 + 0 + 0 + 0 + 0 = 1 \end{aligned}$$

4.2.4 Determinant of a Triangular Matrix

A triangular matrix is a special kind of square matrix in which either entries above or below the main diagonal are zero.

Types of Triangular Matrix

1. Upper Triangular Matrix: All entries below the main diagonal are zero

$$U = \begin{bmatrix} 2 & -6 & 1 \\ 0 & 7 & 3 \\ 0 & 0 & 2 \end{bmatrix}$$

2. Lower Triangular Matrix: All entries above the main diagonal are zero.

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 6 & 0 \\ 6 & 3 & 2 \end{bmatrix}$$

Properties

- The determinant of a triangular matrix is the product of its diagonal elements

$$\det(A) = a_{11} \cdot a_{22} \cdot (\cdots a_{nn})$$

- Triangular matrices are used in LU decomposition and solving linear systems.

Proof

Let $U = [u_{ij}]$ be an $n \times n$ **upper triangular matrix**, so that:

$$u_{ij} = 0 \quad \text{whenever } i > j$$

From the general determinant formula:

$$\det(U) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot u_{1\sigma(1)} u_{2\sigma(2)} \cdots u_{n\sigma(n)}$$

In an upper triangular matrix, if $\sigma(i) < i$ for any i, then $a_{i\sigma(i)} = 0$, and that term contributes nothing.

Only the identity permutation $\sigma(i) = i$ for all i satisfies $\sigma(i) \geq i$ for all i. Hence,

the only surviving term in the sum is:

$$\det(U) = \text{sgn}(\text{id}) \cdot u_{11}u_{22}\cdots u_{nn} = 1 \cdot \prod_{i=1}^{n} u_{ii}$$

conversely,
In a lower triangular matrix, let $L = [\ell_{ij}]$ be an $n \times n$ **lower triangular matrix**, so that:

$$\ell_{ij} = 0 \quad \text{whenever } i < j$$

From the general determinant formula:

$$\det(L) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \cdot \ell_{1\sigma(1)}\ell_{2\sigma(2)}\cdots\ell_{n\sigma(n)}$$

In a lower triangular matrix, if $\sigma(i) > i$ for any i, then $\ell_{i\sigma(i)} = 0$, and that term contributes nothing. **Only the identity permutation** $\sigma(i) = i$ for all i avoids these zero entries.

Thus:

$$\det(L) = \text{sgn}(\text{id}) \cdot \ell_{11}\ell_{22}\cdots\ell_{nn} = \prod_{i=1}^{n} \ell_{ii}$$

Therefore: The determinant of a triangular matrix is the product of its diagonal elements.

Example:

Let

$$U = \begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\det(U) = 3 \times 4 \times 6 = \boxed{72}$$

4.2.5 Minors

Given a matrix $M \in M_{m\times n}$, not necessarily a square one ($m \neq n$), and for $p \leq m, n$, a $p \times p$ matrix $M^{'} \in M_p$ by retaining only p rows and p columns of M. The determinant of such a matrix $M^{'}$ is called a minor of order p. Once the row indices $i_1 < \cdots < i_p$ and column indices $j_1 < \cdots < j_p$ has been selected, we denote by

$$M \begin{pmatrix} i_1 & i_2 & \cdots i_p \\ j_1 & j_2 & \cdots j_p \end{pmatrix}$$

the corresponding minor. A principal minor is a minor with equal row and column indices, that is, of the form

$$M \begin{pmatrix} i_1 & i_2 & \cdots i_p \\ i_1 & i_2 & \cdots i_p \end{pmatrix}$$

The leading principal minor of order p is

$$M \begin{pmatrix} 1 & 2 & \cdots p \\ 1 & 2 & \cdots p \end{pmatrix}$$

.

Example 1:

Given

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 1 & 5 \\ 6 & 2 & 0 \end{bmatrix}$$

We want to find the minor of the element $a_{12} = 3$.

To find the minor M_{12}, we delete row 1 and column 2:

$$M_{12} = \begin{vmatrix} 4 & 5 \\ 6 & 0 \end{vmatrix} = 4 \cdot 0 - 5 \cdot 6 = -30$$

So the minor $M_{12} = -30$

Example 2: Finding a minor of order 2

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 0 & 1 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

Which means:

- Select 2 rows and 2 columns,
- Form resulting 2×2 matrix,
- Take its determinant.

We choose:

- Rows: 1 and 3
- Columns: 3 and 4

So the matrix is:

$$M \begin{pmatrix} 1 & 3 \\ 3 & 4 \end{pmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

We find the determinant

$$M \begin{pmatrix} 1 & 3 \\ 3 & 4 \end{pmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = (2)(3) - (1)(4) = 6 - 4 = 2$$

So the minor is 2

4.2.6 Cofactor

Let $A \in \mathbb{M}_n(F)$ be a square matrix of order n, and let $i, j \in \{1, 2, \ldots, n\}$.

- The **minor** of the entry a_{ij} is the determinant of the submatrix obtained by removing the i-th row and j-th column of A. It is denoted by:

$$A \begin{pmatrix} \hat{i} \\ \hat{j} \end{pmatrix}$$

  where $\hat{i}$ means all rows except row i, and $\hat{j}$ means all columns except column j.

- The **cofactor** c_{ij} of the entry a_{ij} is defined as:

$$c_{ij} := (-1)^{i+j} \cdot \det A \begin{pmatrix} \hat{i} \\ \hat{j} \end{pmatrix}$$

- The **cofactor matrix** C is the matrix whose (i, j)-th entry is the cofactor c_{ij}:

$$C = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}$$

Example

Let

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix} \qquad \cdots \qquad (4.2.6.1)$$

We compute the cofactor $C_{ij} = (-1)^{i+j} \cdot \det(A_{ij})$ for each element.

First Row

$$c_{11} = \boxed{2} \qquad c_{12} = \boxed{10} \qquad c_{13} = \boxed{-20}$$

Second Row

$$c_{21} = \boxed{-1} \qquad c_{22} = \boxed{-5} \qquad c_{23} = \boxed{17}$$

Third Row

$$c_{31} = \boxed{2} \qquad c_{32} = \boxed{-4} \qquad c_{33} = \boxed{8}$$

Cofactor Matrix

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} = \begin{bmatrix} 2 & 10 & -20 \\ -1 & -5 & 17 \\ 2 & -4 & 8 \end{bmatrix}$$

4.2.7 Adjugate

The adjugate matrix (or adjoint), denoted $\text{adj}(M)$, is the transpose of the cofactor matrix:

$$\text{adj}(A) := C^T$$

Example

Compute the Adjugate Matrix

$$\text{adj}(A) = C^T = \begin{bmatrix} 2 & -1 & 2 \\ 10 & -5 & -4 \\ -20 & 17 & 8 \end{bmatrix}$$

4.2.8 The Cauchy Binet Formula

Let A be an $n \times m$ matrix and B be an $m \times n$ matrix, with $n \leq m$. Then the determinant of the $n \times n$ product matrix AB is given by:

$$\det(AB) = \sum_{S \subseteq \{1,2,\ldots,m\}, |S|=n} \det(A_S) \cdot \det(B^S)$$

where:

- S is a subset of $\{1, 2, \ldots, m\}$ with n elements.
- A_S is the $n \times n$ submatrix of A consisting of the columns indexed by S.
- B^S is the $n \times n$ submatrix of B consisting of the rows indexed by S.

4.3 INVERTIBILITY

4.3.1 Definition:

Let $A \in \mathbb{M}_n(R)$, the set of all $n \times n$ matrices over a commutative ring R. We say that A is **invertible** (or **non-singular**) if there exists a matrix $B \in \mathbb{M}_n(R)$ such that:

$$AB = BA = I_n$$

where I_n is the $n \times n$ identity matrix. The matrix B is then called the **inverse** of A, and is denoted by A^{-1}.

4.3.2 Properties of Invertible Matrices

1. An $n \times n$ matrix A has an inverse if and only if its determinant is invertible in R
2. The inverse of a matrix, if it exists, is unique.
3. If A and B are invertible, then AB is also invertible and:

$$(AB)^{-1} = B^{-1}A^{-1}$$

4. The transpose of an invertible matrix is also invertible, and:

$$(A^T)^{-1} = (A^{-1})^T$$

Invertibility and Determinant over a Commutative Ring (Property 1)

Theorem:
Let $A \in M_n(R)$, where R is a commutative ring. Then A is invertible (i.e., there exists $B \in M_n(R)$ such that $AB = BA = I_n$) if and only if $\det(A)$ is a unit in R.
Proof
Suppose A is invertible. Then there exists $B \in M_n(R)$ such that:

$$AB = I_n$$

Taking the determinant of both sides:

$$\det(AB) = \det(I_n)$$

$$\det(A)\det(B) = 1$$

This means that $\det(A)$ has a multiplicative inverse in R, so $\det(A)$ is a unit in R.
Suppose $\det(A)$ is a unit in R. Then we define the inverse of A using the classical adjugate formula:

$$A^{-1} = \frac{1}{\det(A)} \cdot \operatorname{adj}(A)$$

Since $\det(A)$ is a unit, its inverse exists in R, and $\operatorname{adj}(A)$ has entries in R. Therefore, A^{-1} also has entries in R, and:

$$AA^{-1} = A\left(\frac{1}{\det(A)}\operatorname{adj}(A)\right) = I_n$$

Hence, A is invertible in $M_n(R)$.
Conclusion: A matrix over a commutative ring R is invertible if and only if its determinant is a unit in R.

Example:
Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Then:

$$\det(A) = (1)(4) - (2)(3) = 4 - 6 = -2$$

In $\mathbb{Z}$, only ± 1 are invertible. So -2 is not a unit, hence A is not invertible over $\mathbb{Z}$, even though it is invertible over $\mathbb{Q}$ or $\mathbb{R}$.

Uniqueness (Property 2)

Theorem: If a square matrix A has an inverse, then it is unique.

Proof

Suppose A is an $n \times n$ matrix and there exist two matrices B and C such that:

$$AB = BA = I \quad \text{and} \quad AC = CA = I$$

We want to show that $B = C$.

$$B = BI = B(AC) = (BA)C = IC = C$$

Hence, $B = C$.

Conclusion: The inverse of a matrix, if it exists, is unique.

Example:

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

We want to find the inverse A^{-1}, and verify that it is **unique**.

Step 1: Compute A^{-1}

For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \text{if } \det(A) \neq 0,$$

then the inverse is given by:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Here, $a = 1, b = 2, c = 3, d = 4$, so:

$$\det(A) = (1)(4) - (2)(3) = 4 - 6 = -2 \neq 0$$

So,

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

Step 2: Suppose Another Inverse B exists

Assume:

$$B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

and that:

$$AB = I \quad \text{and} \quad BA = I$$

Then:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1x + 2z & 1y + 2w \\ 3x + 4z & 3y + 4w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Solving the system of equations:

$$\begin{cases} x + 2z = 1 \\ y + 2w = 0 \\ 3x + 4z = 0 \\ 3y + 4w = 1 \end{cases}$$

Solving gives:

$$x = -2, \quad z = 1, \quad y = 1.5, \quad w = -0.5$$

So:

$$B = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

Which matches the previous computation of A^{-1}.

Conclusion: No matter how we compute it, the inverse matrix is the same. Hence, the inverse is **unique**.

The Inverse of a Product of Invertible Matrices (Property 3)

Theorem: Let A and B be invertible $n \times n$ matrices over a commutative ring with identity. Then the product AB is invertible and:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Proof

To prove that $(AB)^{-1} = B^{-1}A^{-1}$, we need to show that:

$$(AB)(B^{-1}A^{-1}) = I \quad \text{and} \quad (B^{-1}A^{-1})(AB) = I$$

Left Multiplication:

$$\begin{aligned} (AB)(B^{-1}A^{-1}) &= A(BB^{-1})A^{-1} \\ &= AIA^{-1} \\ &= AA^{-1} \\ &= I \end{aligned}$$

Right Multiplication:

$$\begin{aligned}(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}IB \\ &= B^{-1}B \\ &= I\end{aligned}$$

Conclusion: Since both left and right products yield the identity matrix, we conclude that:

$$(AB)^{-1} = B^{-1}A^{-1}$$

The Transpose of an Invertible Matrix (Property 4)

Theorem: If A is an invertible matrix, then its transpose A^T is also invertible, and:

$$(A^T)^{-1} = (A^{-1})^T$$

Proof

Suppose A is invertible, so that:

$$AA^{-1} = I$$

Take the transpose of both sides:

$$(AA^{-1})^T = I^T = I$$

Using the property of transposes $(AB)^T = B^T A^T$, we get:

$$(A^{-1})^T A^T = I \quad \text{and} \quad A^T(A^{-1})^T = I$$

Thus, $(A^{-1})^T$ is the inverse of A^T, and:

$$(A^T)^{-1} = (A^{-1})^T$$

4.3.3 Computation of Inverse (Adjugate Formula)

If $\det(A) \neq 0$, then the inverse of A can be computed as:

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$$

where $\text{adj}(A)$ is the adjugate matrix, obtained by taking the transpose of the cofactor matrix of A.

Example

Let

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

From Example (4.2.6.1):

$$\det(\text{A}) = 14$$

and

$$\text{adj(A)} = C^T = \begin{bmatrix} 2 & -1 & 2 \\ 10 & -5 & -4 \\ -20 & 17 & 8 \end{bmatrix}$$

So:

$$A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A) = \frac{1}{14} \cdot \begin{bmatrix} 2 & -1 & 2 \\ 10 & -5 & -4 \\ -20 & 17 & 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{-1}{14} & \frac{1}{7} \\ \frac{5}{7} & \frac{-5}{14} & \frac{-2}{7} \\ \frac{-10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

4.3.4 Gauss-Jordan Elimination Method

This method uses elementary row operations to reduce a matrix to the identity matrix, while applying the same operations to the identity matrix. The result on the right becomes the inverse. We form an augmented matrix

$$\left[A|I_n\right]$$

where I_n is $n \times n$ identity matrix.

We then apply rows operations to convert the left of A into the identity matrix I_n.

Finally, the right side of the augmented matrix will become A^{-1}, the inverse of A.

Example:

Given:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

We augment with the identity matrix:

$$\left[\begin{array}{ccc|ccc} 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 4 & 2 & 0 & 1 & 0 \\ 5 & -1 & 0 & 0 & 0 & 1 \end{array}\right]$$

Step 1: Make leading 1 in row 1 by dividing R_1 by 2:

$$R_1 \to \frac{1}{2}R_1 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.5 & 0.5 & 0.5 & 0 & 0 \\ 0 & 4 & 2 & 0 & 1 & 0 \\ 5 & -1 & 0 & 0 & 0 & 1 \end{array}\right]$$

continuing the process...

We obtain:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ 0 & 1 & 0 & \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ 0 & 0 & 1 & -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{array}\right]$$

Thus, the inverse of A is:

$$A^{-1} = \begin{bmatrix} \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

4.3.5 LU Decomposition Method

The LU decomposition method is used to decompose a square matrix into the product of a lower triangular matrix (L) and upper triangular matrix (U).
Given a square matrix A, LU decomposition finds two matrices:

$$A = LU$$

To find the inverse of a matrix A, denoted A^{-1}, it is important to solve:

$$AA^{-1} = I$$

Where I is the identity matrix.
Using LU decomposition:

1. Decompose $A = LU$
2. For each column i of the identity matrix I:
   - Solve $Ly = b_i$ using forward substitution
   - Then solve $Ux = y$ using backward substitution
   - The result x is the ith column of A^{-1}
3. Repeat this for each column of I to get the full inverse

Example: Given the matrix:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

We want to compute A^{-1} using LU decomposition. That is, we first write:

$$A = LU$$

Where L is a lower triangular matrix with 1s on the diagonal, and U is an upper triangular matrix.

Step 1: We perform LU Decomposition

We apply Gaussian elimination to reduce A to upper triangular form U, while recording the multipliers in L.

Start with:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

To eliminate below the pivot in column 1: Row 3: $R_3 \leftarrow R_3 - \frac{5}{2}R_1$

$$R_3 = [5, -1, 0] - \frac{5}{2} \cdot [2, 3, 1] = [0, -1 - \frac{15}{2}, 0 - \frac{5}{2}] = [0, -\frac{17}{2}, -\frac{5}{2}]$$

Now the matrix becomes:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & -\frac{17}{2} & -\frac{5}{2} \end{bmatrix}$$

No need to eliminate anything in row 2 since it's already triangular.

Next, Row 3: $R_3 \leftarrow R_3 + \frac{17}{8}R_2$

$$R_3 = [0, \frac{-17}{2}, \frac{-5}{2}] + \frac{17}{8} \cdot [0, 4, 2] = [0, \frac{-17}{2} + \frac{17}{2}, \frac{-5}{2} + \frac{17}{4}] = [0, 0, \frac{7}{4}]$$

Now the matrix becomes:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}$$

Now we fill in L using the multipliers: $l_{31} = \frac{5}{2}$, $l_{21} = 0$, $l_{32} = -\frac{17}{8}$

So:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{5}{2} & -\frac{17}{8} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}$$

Step 2: We solve $AX = I$ using LU

Let $AX = I \Rightarrow LUX = I$

Let $UX = Y$, so we first solve $LY = I$, then $UX = Y$

We solve 3 systems: $AX_1 = e_1$, $AX_2 = e_2$, $AX_3 = e_3$

Solving for the first column of A^{-1}:

1. We let $LY = e_1 = [1\ 0\ 0]^T$

Forward substitution:

$$L \cdot Y_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} y_1 = 1 \\ y_2 = 0 \\ \frac{5}{2}y_1 - \frac{17}{8}y_2 + y_3 = 0 \Rightarrow y_3 = -\frac{5}{2} \end{cases} \Rightarrow Y_1 = \begin{bmatrix} 1 \\ 0 \\ -\frac{5}{2} \end{bmatrix}$$

2. We solve $UX_1 = Y_1$

Back substitution:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}, \quad UX_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow \begin{cases} \frac{7}{4}x_3 = -\frac{5}{2} \Rightarrow x_3 = -\frac{10}{7} \\ 4x_2 + 2x_3 = 0 \Rightarrow x_2 = \frac{5}{7} \\ 2x_1 + 3x_2 + x_3 = 1 \Rightarrow x_1 = \frac{1}{7} \end{cases}$$

So first column of A^{-1} is:

$$X_1 = \begin{bmatrix} \frac{1}{7} \\ \frac{5}{7} \\ -\frac{10}{7} \end{bmatrix}$$

For column 2 of A^{-1}:

1. We let $LY = e_2 = [0\ 1\ 0]^T$

Forward substitution:

$$L \cdot Y_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} y_1 = 0 \\ y_2 = 1 \\ \frac{5}{2}y_1 - \frac{17}{8}y_2 + y_3 = 0 \Rightarrow y_3 = \frac{17}{8} \end{cases} \Rightarrow Y_2 = \begin{bmatrix} 0 \\ 1 \\ \frac{17}{8} \end{bmatrix}$$

2. Now we solve $UX_2 = Y_2$

Back substitution:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}, \quad UX_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow \begin{cases} \frac{7}{4}x_3 = \frac{17}{8} \Rightarrow x_3 = \frac{17}{14} \\ 4x_2 + 2x_3 = 1 \Rightarrow x_2 = -\frac{5}{14} \\ 2x_1 + 3x_2 + x_3 = 0 \Rightarrow x_1 = -\frac{1}{14} \end{cases}$$

Thus second column of A^{-1} is:

$$X_2 = \begin{bmatrix} -\frac{1}{14} \\ -\frac{5}{14} \\ \frac{17}{14} \end{bmatrix}$$

For column 3 of A^{-1}:

1. We let $LY = e_3 = [0\ 0\ 1]^T$

Forward substitution:

$$L \cdot Y_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{cases} y_1 = 0 \\ y_2 = 0 \\ \frac{5}{2}y_1 - \frac{17}{8}y_2 + y_3 = 1 \Rightarrow y_3 = 1 \end{cases} \Rightarrow Y_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

2. Now we solve $UX_3 = Y_3$

Back substitution:

$$U = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & \frac{7}{4} \end{bmatrix}, \quad UX_3 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow \begin{cases} \frac{7}{4}x_3 = 1 \Rightarrow x_3 = \frac{4}{7} \\ 4x_2 + 2x_3 = 0 \Rightarrow x_2 = -\frac{2}{7} \\ 2x_1 + 3x_2 + x_3 = 0 \Rightarrow x_1 = \frac{1}{7} \end{cases}$$

Thus third column of A^{-1} is:

$$X_3 = \begin{bmatrix} \frac{1}{7} \\ -\frac{2}{7} \\ \frac{4}{7} \end{bmatrix}$$

Final Result:

$$A^{-1} = \begin{bmatrix} -\frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

4.3.6 Block Matrix Method for Matrix Inversion

Let

$$A = \begin{bmatrix} P & Q \\ R & S \end{bmatrix}$$

be a square matrix partitioned into four blocks, where:

- P is invertible,
- The **Schur complement** of P in A, defined by $S - RP^{-1}Q$, is also invertible.

Then, the inverse of A is given by:

$$A^{-1} = \begin{bmatrix} P^{-1} + P^{-1}Q(S - RP^{-1}Q)^{-1}RP^{-1} & -P^{-1}Q(S - RP^{-1}Q)^{-1} \\ -(S - RP^{-1}Q)^{-1}RP^{-1} & (S - RP^{-1}Q)^{-1} \end{bmatrix}$$

Notice:

- The matrix $S - RP^{-1}Q$ is called the **Schur complement** of P in A.
- The formula can also be adapted when S is invertible instead of P, using the Schur complement of S in A, i.e., $P - QS^{-1}R$.

Example:

Let the matrix A be given by:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

We partition A as:

$$A = \begin{bmatrix} P & Q \\ R & S \end{bmatrix} = \left[\begin{array}{cc|c} 2 & 3 & 1 \\ 0 & 4 & 2 \\ \hline 5 & -1 & 0 \end{array}\right]$$

Define the blocks:

$$P = \begin{bmatrix} 2 & 3 \\ 0 & 4 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad R = \begin{bmatrix} 5 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 0 \end{bmatrix}$$

Step 1: Compute P^{-1}

$$\det(P) = 2 \cdot 4 - 0 \cdot 3 = 8, \quad P^{-1} = \frac{1}{8}\begin{bmatrix} 4 & -3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{3}{8} \\ 0 & \frac{1}{4} \end{bmatrix}$$

Step 2: Compute Schur complement: $S - RP^{-1}Q$

$$RP^{-1} = \begin{bmatrix} 5 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{3}{8} \\ 0 & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{5}{2} & -\frac{17}{8} \end{bmatrix}$$

$$RP^{-1}Q = \begin{bmatrix} \frac{5}{2} & -\frac{17}{8} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{5}{2} - \frac{34}{8} = -\frac{7}{4} \Rightarrow (S - RP^{-1}Q)^{-1} = \left(\frac{7}{4}\right)^{-1} = \frac{4}{7}$$

Step 3: Compute $P^{-1}Q$

$$P^{-1}Q = \begin{bmatrix} \frac{1}{2} & -\frac{3}{8} \\ 0 & \frac{1}{4} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{2} \end{bmatrix} \Rightarrow P^{-1}Q(S - RP^{-1}Q)^{-1} = \frac{4}{7} \begin{bmatrix} -\frac{1}{4} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{7} \\ \frac{2}{7} \end{bmatrix}$$

Step 4: Compute full inverse using block formula

$$P^{-1}Q(S - RP^{-1}Q)^{-1}RP^{-1} = \begin{bmatrix} -\frac{1}{7} \\ \frac{2}{7} \end{bmatrix} \begin{bmatrix} \frac{5}{2} & -\frac{17}{8} \end{bmatrix} = \begin{bmatrix} -\frac{5}{14} & \frac{17}{56} \\ \frac{5}{7} & -\frac{17}{28} \end{bmatrix}$$

$$\Rightarrow \text{Top-left block: } \begin{bmatrix} \frac{1}{2} & -\frac{3}{8} \\ 0 & \frac{1}{4} \end{bmatrix} + \begin{bmatrix} -\frac{5}{14} & \frac{17}{56} \\ \frac{5}{7} & -\frac{17}{28} \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & -\frac{1}{14} \\ \frac{5}{7} & -\frac{5}{14} \end{bmatrix}$$

$$\text{Top-right block: } -P^{-1}Q(S - RP^{-1}Q)^{-1} = \begin{bmatrix} \frac{1}{7} \\ -\frac{2}{7} \end{bmatrix}$$

$$\text{Bottom-left block: } -\frac{4}{7} \begin{bmatrix} \frac{5}{2} & -\frac{17}{8} \end{bmatrix} = \begin{bmatrix} -\frac{10}{7} & \frac{17}{14} \end{bmatrix}$$

$$\text{Bottom-right block: } \frac{4}{7}$$

Final inverse:

$$A^{-1} = \begin{bmatrix} \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

4.3.7 Cayley-Hamilton Theorem Method

Definition

The **Cayley-Hamilton Theorem** states that every square matrix satisfies its own characteristic equation.

Let $A \in M_n(F)$, the set of all $n \times n$ matrices over a field F, and let its characteristic polynomial be:

$$\chi_A(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

Then, by the Cayley-Hamilton Theorem:

$$A^n + a_{n-1}A^{n-1} + \cdots + a_1A + a_0I = 0$$

Using the Theorem to Find A^{-1}

If A is invertible, then the constant term $a_0 \neq 0$. We can rearrange the equation to express A^{-1} as:

$$A^{-1} = -\frac{1}{a_0}(A^{n-1} + a_{n-1}A^{n-2} + \cdots + a_1I)$$

Example

Let:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

Find the Characteristic Polynomial

$$\det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 3 & 1 \\ 0 & 4-\lambda & 2 \\ 5 & -1 & -\lambda \end{bmatrix}$$

Using cofactor expansion along the first row:

$$= (2-\lambda) \cdot \begin{vmatrix} 4-\lambda & 2 \\ -1 & -\lambda \end{vmatrix} - 3 \cdot \begin{vmatrix} 0 & 2 \\ 5 & -\lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & 4-\lambda \\ 5 & -1 \end{vmatrix}$$

Now compute each 2×2 determinant:

$$\begin{vmatrix} 4-\lambda & 2 \\ -1 & -\lambda \end{vmatrix} = (4-\lambda)(-\lambda) - (-1)(2)$$

$$= -4\lambda + \lambda^2 + 2 = \lambda^2 - 4\lambda + 2$$

$$\begin{vmatrix} 0 & 2 \\ 5 & -\lambda \end{vmatrix} = 0 \cdot (-\lambda) - 5 \cdot 2 = -10$$

$$\begin{vmatrix} 0 & 4-\lambda \\ 5 & -1 \end{vmatrix} = 0 \cdot (-1) - 5 \cdot (4-\lambda) = -20 + 5\lambda$$

Putting it together:

$$\begin{aligned} \det(A - \lambda I) &= (2-\lambda)(\lambda^2 - 4\lambda + 2) + 30 + (5\lambda - 20) \\ &= (2-\lambda)(\lambda^2 - 4\lambda + 2) + 5\lambda + 10 \end{aligned}$$

Now expand:

$$\begin{aligned} (2-\lambda)(\lambda^2 - 4\lambda + 2) &= 2\lambda^2 - 8\lambda + 4 - (\lambda^3 - 4\lambda^2 + 2\lambda) \\ &= -\lambda^3 + 6\lambda^2 - 10\lambda + 4 \end{aligned}$$

So:

$$\begin{aligned} \det(A - \lambda I) &= -\lambda^3 + 6\lambda^2 - 10\lambda + 4 + 5\lambda + 10 \\ &= -\lambda^3 + 6\lambda^2 - 5\lambda + 14 \end{aligned}$$

Hence, the characteristic polynomial is:

$$p(\lambda) = -\lambda^3 + 6\lambda^2 - 5\lambda + 14$$

By the Cayley-Hamilton Theorem:

$$A^3 - 6A^2 + 5A - 14I = 0$$

Multiply both sides by A^{-1}:

$$A^2 - 6A + 5I = 14A^{-1} \Rightarrow A^{-1} = \frac{1}{14}(A^2 - 6A + 5I)$$

Compute A^2

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

$$A^2 = A \cdot A = \begin{bmatrix} 9 & 17 & 8 \\ 10 & 14 & 8 \\ 10 & 11 & 3 \end{bmatrix}$$

Compute $A^2 - 6A + 5I$

$$6A = \begin{bmatrix} 12 & 18 & 6 \\ 0 & 24 & 12 \\ 30 & -6 & 0 \end{bmatrix}, \quad 5I = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Now compute:

$$A^2 - 6A + 5I = \begin{bmatrix} 9 & 17 & 8 \\ 10 & 14 & 8 \\ 10 & 11 & 3 \end{bmatrix} - \begin{bmatrix} 12 & 18 & 6 \\ 0 & 24 & 12 \\ 30 & -6 & 0 \end{bmatrix} + \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Compute entry-wise:

$$= \begin{bmatrix} (9-12+5) & (17-18+0) & (8-6+0) \\ (10-0+0) & (14-24+5) & (8-12+0) \\ (10-30+0) & (11+6+0) & (3-0+5) \end{bmatrix} = \begin{bmatrix} 2 & -1 & 2 \\ 10 & -5 & -4 \\ -20 & 17 & 8 \end{bmatrix}$$

Now:

$$A^{-1} = \frac{1}{14} \begin{bmatrix} 2 & -1 & 2 \\ 10 & -5 & -4 \\ -20 & 17 & 8 \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

Hence we have computed the inverse of A using the Cayley-Hamilton theorem:

$$A^{-1} = \frac{1}{14}(A^2 - 6A + 5I)$$

4.4 Applications of Determinants and Matrix Invertibility

4.4.1 Introduction

Determinants and invertibility of matrices play critical roles across science, engineering, economics, and computer science. Their applications are not just theoretical; they ensure that real-world systems are stable, solvable, and interpretable.

4.4.2 Fundamental concepts

A matrix $A \in M_n(R)$, the set of all $n \times n$ matrices over a commutative ring R, is said to be **invertible** (or **non-singular**) if and only if its determinant $\det(A)$ is an *invertible element* in R (or $\det(A) \neq 0$ if R is a field F). The determinant serves as a critical test for the linear independence of the rows (or columns) of the matrix. If $\det(A) \neq 0$, then the following hold:

- The matrix A has **full rank** (i.e., rank $= n$),
- The system $A\vec{x} = \vec{b}$ has a **unique solution** for every $\vec{b} \in R^n$,
- The linear transformation represented by A is **bijective** (i.e., one-to-one and onto, hence reversible).

4.4.3 Real World Applications

1. Engineering and Computer Graphics

Application: Transformations in 2D and 3D graphics (e.g., rotation, scaling, translation).
Determinant Insight: The determinant of a transformation matrix reflects the area or volume scaling. A determinant of zero indicates loss of dimensionality (e.g., flattening).
Invertibility Use: To reverse transformations, the matrix must be invertible.

2. Data Science and Machine Learning

Application: Algorithms like linear regression, Principal Component Analysis (PCA), and multivariate statistics.
Determinant Insight: The determinant of the covariance matrix reflects data spread. A zero (or near-zero) value signals linear dependence.
Invertibility Use: Inverse matrices are needed to solve normal equations and compute model parameters.

3. Economics and Optimization

Application: Solving input-output economic models or optimization problems.
Determinant Insight: A non-zero determinant of the coefficient matrix indicates a unique solution exists.
Invertibility Use: Enables explicit solutions using $A^{-1}b$.

4. Network Systems and Electrical Circuits

Application: Modeling electrical networks using Kirchhoff's laws.
Determinant Insight: Helps determine the existence and uniqueness of current and voltage solutions.
Invertibility Use: Essential for simulating circuits and solving node equations.

5. Cryptography

Application: Hill cipher and other matrix-based encryption schemes.
Determinant Insight: The key matrix must have a non-zero determinant mod n for decryption to be possible.
Invertibility Use: Without an inverse matrix, decryption fails.

6. Control Systems and Robotics

Application: State-space control, robotic arm movement.
Determinant Insight: Indicates controllability and observability of system matrices.
Invertibility Use: Allows for feedback control design and inverse kinematics.

7. Computer-Aided Design and Simulation

Application: Finite Element Analysis (FEA), structural simulation.
Determinant Insight: Ensures mesh elements are valid (non-inverted).
Invertibility Use: Used to invert stiffness matrices for solving physical systems.

8. Artificial Intelligence and Neuroscience

Application: Neural network modeling and brain connectivity analysis.
Determinant Insight: Reveals synchronization and stability of networks.
Invertibility Use: Needed to reconstruct inputs or stabilize outputs through feedback.

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary of Findings

This study has provided a comprehensive exploration into the interplay between matrix invertibility and determinants, a foundational topic in linear algebra with far-reaching implications across theoretical and applied disciplines. The initial phase of the study focused on establishing fundamental ideas in linear algebra, covering matrix structure, operations, and transformations, and then transitioned into a detailed examination of determinants and their significance. Using both theoretical frameworks and worked-out examples, the study demonstrated that the determinant is a decisive criterion for matrix invertibility. That is, a square matrix is invertible if and only if its determinant is non-zero. Through rigorous analytical treatment and computational demonstrations, several critical insights emerged:

- Determinant as the Core Criterion for Invertibility: The determinant of a square matrix was confirmed as a decisive indicator of whether the matrix is invertible. Specifically, a non-zero determinant was shown to guarantee the existence of a unique inverse, while a zero determinant signals singularity and non-invertibility.

- Structural Properties of Determinants: The research systematically examined key determinant properties; linearity, alternation, and normalization, and established how these properties underpin the computation and theoretical behaviour of matrix inverses.

- Computational Techniques and Verification: Methods such as the adjugate approach, LU decomposition, Gauss-Jordan elimination, and block matrix methods were employed to compute matrix inverses. The study also highlighted the practicality of using software tools for complex calculations.

- Applications of Invertibility: Theoretical and computational findings were contextualized through real-world applications, particularly in cryptography, where invertibility ensures secure encoding and decoding, and in engineering and numerical analysis, where invertible matrices are pivotal in solving systems of equations.

Overall, the project emphasized that determinants provide an efficient and conceptually clear method for evaluating matrix invertibility both in theory and application.

5.2 Conclusions

The determinant stands not merely as a computational tool, but as a conceptual anchor for understanding matrix invertibility. Its evaluation yields immediate insight into the solvability of linear systems and the behaviour of linear transformations. This work underscores that:

- The condition $\det(A) \neq 0$ is both necessary and sufficient for the invertibility of square matrices in standard fields $(\mathbb{R}, \mathbb{C})$
- The determinant's properties extend beyond arithmetic utility, they reflect deeper algebraic and geometric relationships within the matrix structure.
- Classical methods of computing the inverse are still relevant, yet modern computational techniques significantly enhance the accuracy and scalability of such tasks.

In sum, the determinant serves as a bridge between abstract algebraic theory and concrete applied computation.

5.3 Recommendations

Informed by the study's results, these actions are recommended:

- Educational Integration: Linear algebra curricula should emphasize the conceptual linkage between determinants and invertibility, moving beyond procedural computation to include interpretative understanding.
- Tool-Assisted Learning: Instructors and learners should incorporate mathematical software (e.g., MATLAB, Mathematica, Python's NumPy) to explore determinant-based invertibility in dynamic ways.
- Cross-Disciplinary Applications: Researchers in cryptography, data science, and engineering should actively apply determinant criteria in algorithm design and system modelling, especially in contexts requiring reversible operations.

Future Research Directions:

- Generalized Inverses and Non-Square Matrices: While this study focused on square matrices over standard fields, future work could investigate determinant-like conditions in generalized inverse frameworks, such as the Moore–Penrose and core inverse over rings and inclines.
- Invertibility in Abstract Structures: Extending the analysis to matrices defined over non-commutative rings or modules would reveal deeper algebraic phenomena and widen the applicability of the theory.
- Algorithmic Optimization: Future computational research may explore more efficient determinant-based invertibility tests for high-dimensional or sparse matrices, particularly those relevant to machine learning and quantum computing.

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Appendices

Appendix A: Detailed Workout

A.1 The cofactor expansion for example (4.2.6.1)

Let

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix} \qquad \cdots \qquad (4.2.6.1)$$

We compute the cofactor $C_{ij} = (-1)^{i+j} \cdot \det(A_{ij})$ for each element.

First Row

$$c_{11} = (+1) \cdot \begin{vmatrix} 4 & 2 \\ -1 & 0 \end{vmatrix} = 1 \cdot (4 \cdot 0 - 2 \cdot (-1)) = 1 \cdot (0 + 2) = \boxed{2}$$

$$c_{12} = (-1) \cdot \begin{vmatrix} 0 & 2 \\ 5 & 0 \end{vmatrix} = -1 \cdot (0 \cdot 0 - 2 \cdot 5) = -1 \cdot (-10) = \boxed{10}$$

$$c_{13} = (+1) \cdot \begin{vmatrix} 0 & 4 \\ 5 & -1 \end{vmatrix} = 1 \cdot (0 \cdot (-1) - 4 \cdot 5) = \boxed{-20}$$

Second Row

$$c_{21} = (-1) \cdot \begin{vmatrix} 3 & 1 \\ -1 & 0 \end{vmatrix} = -1 \cdot (3 \cdot 0 - 1 \cdot (-1)) = -1 \cdot (0 + 1) = \boxed{-1}$$

$$c_{22} = (+1) \cdot \begin{vmatrix} 2 & 1 \\ 5 & 0 \end{vmatrix} = 1 \cdot (2 \cdot 0 - 1 \cdot 5) = 1 \cdot (-5) = \boxed{-5}$$

$$c_{23} = (-1) \cdot \begin{vmatrix} 2 & 3 \\ 5 & -1 \end{vmatrix} = -1 \cdot (2 \cdot (-1) - 3 \cdot 5) = -1 \cdot (-2 - 15) = \boxed{17}$$

Third Row

$$c_{31} = (+1) \cdot \begin{vmatrix} 3 & 1 \\ 4 & 2 \end{vmatrix} = 1 \cdot (3 \cdot 2 - 1 \cdot 4) = \boxed{2}$$

$$c_{32} = (-1) \cdot \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = -1 \cdot (2 \cdot 2 - 1 \cdot 0) = \boxed{-4}$$

$$c_{33} = (+1) \cdot \begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix} = 1 \cdot (2 \cdot 4 - 3 \cdot 0) = \boxed{8}$$

Cofactor Matrix

$$C = \begin{bmatrix} 2 & 10 & -20 \\ -1 & -5 & 17 \\ 2 & -4 & 8 \end{bmatrix}$$

A.2 Using Gauss-Jordan Elimination Method to find the inverse of example (4.2.6.1)

Given:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

We augment with the identity matrix:

$$\left[\begin{array}{ccc|ccc} 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 4 & 2 & 0 & 1 & 0 \\ 5 & -1 & 0 & 0 & 0 & 1 \end{array}\right]$$

Step 1: Make leading 1 in row 1 by dividing R_1 by 2:

$$R_1 \to \frac{1}{2}R_1 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.5 & 0.5 & 0.5 & 0 & 0 \\ 0 & 4 & 2 & 0 & 1 & 0 \\ 5 & -1 & 0 & 0 & 0 & 1 \end{array}\right]$$

Step 2: Eliminate below using $R_3 \to R_3 - 5R_1$:

$$\left[\begin{array}{ccc|ccc} 1 & 1.5 & 0.5 & 0.5 & 0 & 0 \\ 0 & 4 & 2 & 0 & 1 & 0 \\ 0 & -8.5 & -2.5 & -2.5 & 0 & 1 \end{array}\right]$$

Step 3: Normalize row 2:

$$R_2 \to \frac{1}{4}R_2 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 1.5 & 0.5 & 0.5 & 0 & 0 \\ 0 & 1 & 0.5 & 0 & 0.25 & 0 \\ 0 & -8.5 & -2.5 & -2.5 & 0 & 1 \end{array}\right]$$

Step 4: Eliminate above and below pivot in column 2:

$$R_1 \to R_1 - 1.5R_2, \quad R_3 \to R_3 + 8.5R_2 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -0.25 & 0.5 & -0.375 & 0 \\ 0 & 1 & 0.5 & 0 & 0.25 & 0 \\ 0 & 0 & 1.75 & -2.5 & 2.125 & 1 \end{array}\right]$$

Step 5: Normalize row 3:

$$R_3 \to \frac{1}{1.75}R_3 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -0.25 & 0.5 & -0.375 & 0 \\ 0 & 1 & 0.5 & 0 & 0.25 & 0 \\ 0 & 0 & 1 & -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{array}\right]$$

Step 6: Eliminate above pivot in column 3:

$$R_1 \to R_1 + 0.25R_3, \quad R_2 \to R_2 - 0.5R_3 \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ 0 & 1 & 0 & \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ 0 & 0 & 1 & -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{array}\right]$$

Thus, the inverse of A is:

$$A^{-1} = \begin{bmatrix} \frac{1}{7} & -\frac{1}{14} & \frac{1}{7} \\ \frac{5}{7} & -\frac{5}{14} & -\frac{2}{7} \\ -\frac{10}{7} & \frac{17}{14} & \frac{4}{7} \end{bmatrix}$$

Appendix B: Proofs of Theorems

B.1 Proof of the General Determinant Formula for a 3×3 Matrix

Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ $\qquad n = 3 \qquad S_3 = 3! = 6$ permutations: $(\sigma_1, \sigma_2, \cdots, \sigma_6)$

Using the formula

$$\varepsilon(\sigma) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i}$$

to find the sign $\varepsilon(\sigma)$ of permutation in S_3.

Using the set {1,2,3}, so all permutation σ of size 3 are;

- $\sigma_1 = (1, 2, 3)$, the identity permutation.

$$\sigma_1(1) = 1, \qquad \sigma_1(2) = 2, \qquad \sigma_1(3) = 3$$

$$\varepsilon(\sigma_1) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i} = \frac{1-2}{1-2} \cdot \frac{1-3}{1-3} \cdot \frac{2-3}{2-3} = \frac{-2}{-2} = +1$$

$\varepsilon(\sigma_1) = +1$

- $\sigma_2 = (1, 2) = (2, 1, 3)$

$$\sigma_2(1) = 2, \qquad \sigma_2(2) = 1, \qquad \sigma_2(3) = 3$$

$$\varepsilon(\sigma_2) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i} = \frac{2-1}{1-2} \cdot \frac{2-3}{1-3} \cdot \frac{1-3}{2-3} = \frac{2}{-2} = -1$$

$\varepsilon(\sigma_2) = -1$

- $\sigma_3 = (1, 3) = (3, 2, 1)$

$$\sigma_3(1) = 3, \qquad \sigma_3(2) = 2, \qquad \sigma_3(3) = 1$$

$$\varepsilon(\sigma_3) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i} = \frac{3-2}{1-2} \cdot \frac{3-1}{1-3} \cdot \frac{2-1}{2-3} = \frac{2}{-2} = -1$$

$\varepsilon(\sigma_3) = -1$

- $\sigma_4 = (2,3) = (1,3,2)$

$$\sigma_4(1) = 1, \qquad \sigma_4(2) = 3, \qquad \sigma_4(3) = 2$$

$$\varepsilon(\sigma_2) = \prod_{i=1}^{3} \frac{\sigma(j) - \sigma(i)}{j - i} = \frac{1-3}{1-2} \cdot \frac{1-2}{1-3} \cdot \frac{3-2}{2-3} = \frac{2}{-2} = -1$$

  $\varepsilon(\sigma_4) = -1$

- $\sigma_5 = (2,3,1)$
  $\varepsilon(\sigma_5) = +1$

- $\sigma_6 = (3,1,2)$
  $\varepsilon(\sigma_6) = +1$

To generate the terms,

- $\sigma_1 = (1,2,3) = a_{1\sigma_1(1)} \cdot a_{2\sigma_1(2)} \cdot a_{3\sigma_1(3)}$
  $= a_{11}a_{22}a_{33}$

- $\sigma_2 = (1,2) = (2,1,3) = a_{1\sigma_2(1)} \cdot a_{2\sigma_2(2)} \cdot a_{3\sigma_2(3)}$
  $= a_{12}a_{21}a_{33}$

- $\sigma_3 = (1,3) = (3,2,1) = a_{1\sigma_3(1)} \cdot a_{2\sigma_3(2)} \cdot a_{3\sigma_3(3)}$
  $= a_{13}a_{22}a_{31}$

- $\sigma_4 = (2,3) = (1,3,2) = a_{1\sigma_4(1)} \cdot a_{2\sigma_4(2)} \cdot a_{3\sigma_4(3)}$
  $= a_{11}a_{23}a_{32}$

- $\sigma_5 = (2,3,1) = a_{1\sigma_5(1)} \cdot a_{2\sigma_5(2)} \cdot a_{3\sigma_5(3)}$
  $= a_{12}a_{23}a_{31}$

- $\sigma_6 = (3,1,2) = a_{1\sigma_6(1)} \cdot a_{2\sigma_6(2)} \cdot a_{3\sigma_6(3)}$
  $= a_{13}a_{21}a_{32}$

Now

$$det(A) = \sum_{\sigma \in S_3} \varepsilon(\sigma_1) \cdot a_{1\sigma_1(1)}a_{2\sigma_1(2)}a_{3\sigma_1(3)} + \varepsilon(\sigma_2) \cdot a_{1\sigma_2(1)}a_{2\sigma_2(2)}a_{3\sigma_1(3)} + \cdots$$

$$+\varepsilon(\sigma_6) \cdot a_{1\sigma_6(1)}a_{2\sigma_6(2)}a_{3\sigma_6(3)}$$

$$= (+1) \cdot a_{11}a_{22}a_{33} + (-1) \cdot a_{12}a_{21}a_{33} + (-1) \cdot a_{13}a_{22}a_{31} + (-1) \cdot a_{11}a_{23}a_{32} +$$

$$(+1) \cdot a_{12}a_{23}a_{31} + (+1) \cdot a_{13}a_{21}a_{32}$$

$$= a_{11}a_{22}a_{33} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$

$$= aei - bdi - ceg - afh + bfg + cdh$$

$$= aei + bfg + cdh - afh - bdi - ceg$$

B.2 Cauchy–Binet Formula

Theorem (Cauchy–Binet). We consider the definition of the determinant of AB using the Leibniz formula:

$$\det(AB) = \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \prod_{i=1}^{m} (AB)_{i,\sigma(i)}.$$

Let $A = [a_{ij}]$ be an $n \times m$ matrix, and $B = [b_{jk}]$ an $m \times n$ matrix. Then the product AB is an $n \times n$ matrix with entries:

$$(AB)_{ik} = \sum_{j=1}^{m} a_{ij} b_{jk}$$

Let us expand $\det(AB)$ using the multilinearity and alternation properties of the determinant. The i-th row of AB is:

$$\sum_{j=1}^{m} a_{ij} (b_{j1}, b_{j2}, \ldots, b_{jn})$$

So each row of AB is a linear combination of the rows of B, with coefficients coming from the corresponding row of A. The determinant of AB becomes a multilinear function in the rows of AB, so we can write:

$$\det(AB) = \sum_{j_1=1}^{m} \cdots \sum_{j_n=1}^{m} a_{1j_1} \cdots a_{nj_n} \cdot \det \begin{bmatrix} b_{j_1 1} & \cdots & b_{j_1 n} \\ \vdots & \ddots & \vdots \\ b_{j_n 1} & \cdots & b_{j_n n} \end{bmatrix}$$

Now, note that if any $j_p = j_q$ for $p \neq q$, then two rows of the above matrix are equal, so the determinant is zero. Thus, we only sum over sequences $(j_1, \ldots, j_n)$ with distinct indices — i.e., permutations of subsets of size n.

Group these terms by unordered subsets $S = \{j_1, \ldots, j_n\} \subseteq \{1, 2, \ldots, m\}$. For each such subset S, the terms correspond to permutations of the rows selected from B. This gives:

$$\det(AB) = \sum_{S \subseteq \{1,2,\ldots,m\}, |S|=n} \det(A_S) \cdot \det(B^S)$$

Here, A_S selects the columns indexed by S, and B^S selects the rows indexed by S, in the same order.

Appendix C: Using Software Tools to Compute Inverse

Example Matrix

Let us consider the matrix:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 5 & -1 & 0 \end{bmatrix}$$

1. Using Python (NumPy)

```
import numpy as np
A = np.array([[2, 3, 1], [0, 4, 2], [5, -1, 0]])
#Compute the inverse
A_inv = np.linalg.inv(A)
print(A_inv)
# Verify identity
print(np.dot(A, A_inv))
```

Note: `np.linalg.inv()` uses LU decomposition internally.

2. Using R Language

```
A <- matrix(c(2, 3, 1, 0, 4, 2, 5, -1, 0), nrow = 3, byrow = TRUE)
# Compute the inverse
A_inv <- solve(A)
print(A_inv)
# Verify identity
A %*% A_inv
```

Note: `solve()` function in R computes the inverse using efficient matrix methods.

3. Using MATLAB / Octave

```
A = [2 3 1; 0 4 2; 5 -1 0];
% Compute the inverse
A_inv = inv(A);
disp(A_inv)
% Verify
disp(A * A_inv)
```